

Population Dynamics

England & Wales, 2001–2020

A Demographic Data Analytics Study

Authored By : Hakeem Ololade Safiriyu

331

Local Authorities

2001–2020

19 Years of Data

14.1%

Total Population Growth

3 Datasets

MYEB1 · MYEB2 · MYEB3

1. Data Cleaning & Integration



MYEB1

68,068 rows

Population estimates
by age & sex
(supporting reference)

MYEB2

60,242 rows

Full demographic
components - births,
deaths, migration
(main dataset)

MYEB3

374 rows

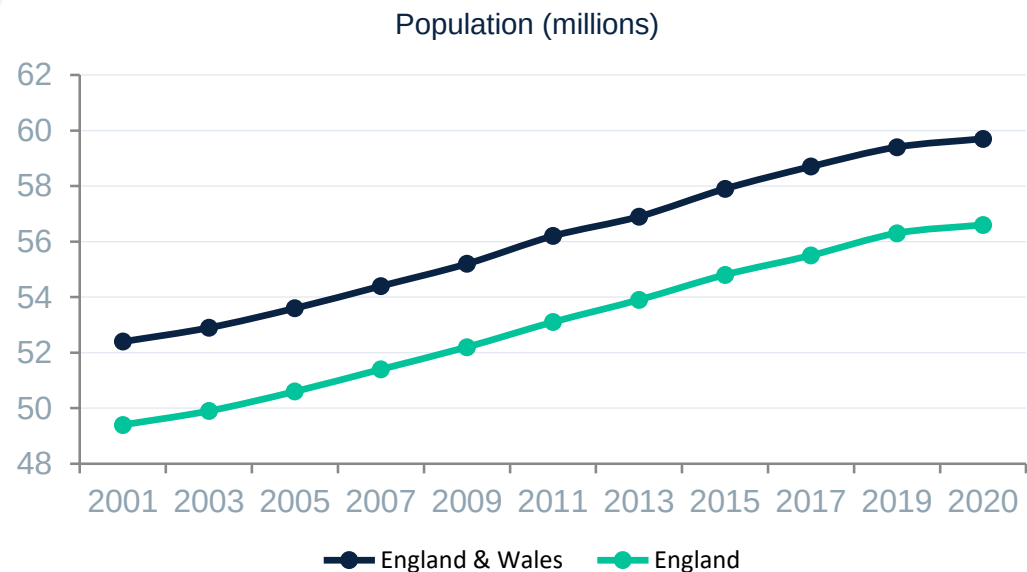
Summary components
(validation benchmark
against MYEB2)

✓ Cross-validation: MYEB2 aggregated vs MYEB3 - zero absolute differences across all 19 years and 5 variables

Key Cleaning Decisions

- Sex recoded: integer codes 1 & 2 → 'Male' / 'Female' using `case_when()`
- Wide → Long reshape: 60,242 × 225 columns → 1,204,840-row panel via `pivot_longer()`
- Year 2001 rate variables set to NA - no prior-year population denominator exists
- Outliers retained: e.g. City of London (48.6% growth, population 7,359) - genuine demographic reality
- Unattrib variable: structural gap from 2012 onward handled using `na.rm = TRUE`

2. Descriptive Analytics – Key Findings



Growth accelerated 2004–2010 following EU expansion — then slowed post-2017

14.1%

Total growth
(England & Wales)

14.4%

England
growth rate

8.9%

Wales
growth rate

56.6%

Growth from net
int'l migration

Demographic Typology – 331 Local Authorities:

182 LAs (55%)

Dual Growth

105 LAs (32%)

Migration Offsets Natural Decline

41 LAs (12%)

Natural Growth Offsets Out-migration

3 LAs (1%)

Dual Decline

3. Hypothesis Testing – 9 Formal Tests



✓ 8 of 9 tests rejected H_0 - Only Test 8 (country vs growth driver, $\chi^2 = 0.221$, $p = 0.638$) was non-significant

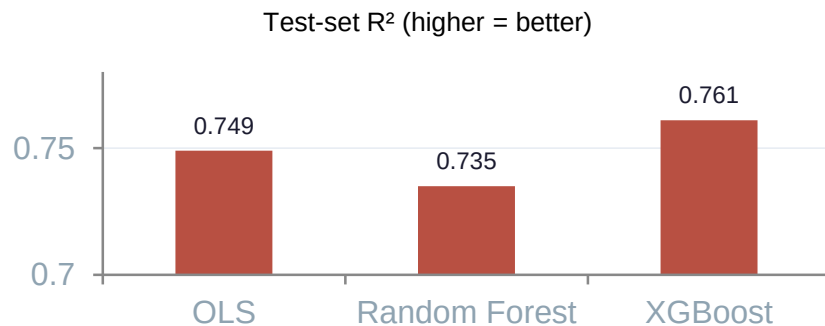
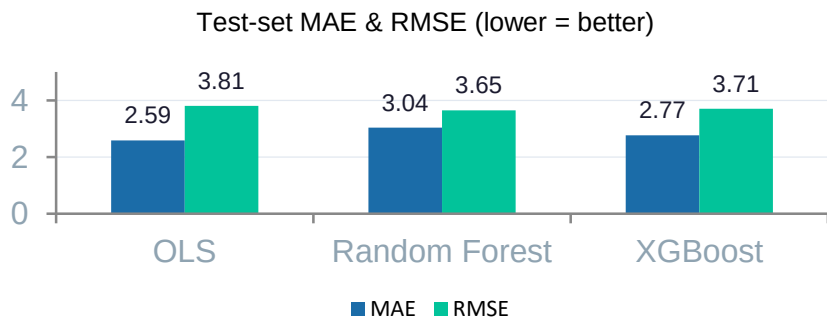
#	Focus	Test	Result	Decision
1	Birth rates: England vs Wales	Welch t-test	$t = 12.75$, $p < 0.001$	Reject H_0
2	Death rates: England vs Wales	Welch t-test	$t = -21.84$, $p < 0.001$	Reject H_0
3	Population totals by sex	Paired t-test	$t = 69.83$, $p < 0.001$	Reject H_0
4	Net migration by sex	Paired t-test	$t = -5.02$, $p < 0.001$	Reject H_0
5	Migration rate across growth classes	ANOVA + Tukey HSD	$F(2,328) = 25.20$, $p < 0.001$	Reject H_0
6	Dependency ratio across growth classes	One-way ANOVA	$F(2,328) = 20.81$, $p < 0.001$	Reject H_0
7	Dependency ratio (non-parametric check)	Kruskal-Wallis	$\chi^2(2) = 36.18$, $p < 0.001$	Reject H_0
8	Country vs main growth driver	Chi-square	$\chi^2(1) = 0.221$, $p = 0.638$	Fail to reject
9	Demographic predictors of growth	Pearson & Spearman	Death $r = -0.535$; Migration $r = 0.483$	Reject H_0

4. Predictive Modelling

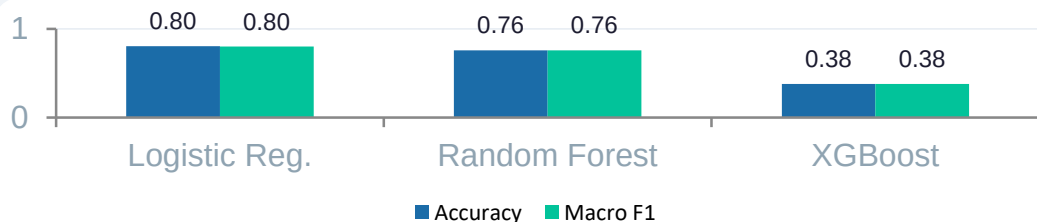


💡 Key modelling insight: migration rate, death rate and birth rate were the strongest recurring predictors across all models

Regression – target: population growth rate 2001–2020



Classification – target: Low / Medium / High growth class



Clustering – k-means, k = 4

Cluster	Label	Growth
n=94	Mixed Profile	17.3%
n=132	Low-growth Suburban	10.1%
n=22	High-growth Urban	26.0%
n=83	Ageing High-Dependency	11.3%

5. Conclusions & Critical Evaluation



Key Findings

- 52.4M → 59.7M total growth (14.1%)
- England grew ~1.6× faster than Wales
- Migration drove 56.6% of total growth
- 8 of 9 hypothesis tests rejected H_0



Best Models

- Regression: OLS best MAE (2.59)
- Clustering: 4 clear demographic typologies
- Classification: Logistic Reg. best (80.3%)
- Migration rate = dominant predictor across all models



Tool Evaluation

- ggplot2: flexible & reproducible, but static output
- tidyverse: clean, modular, readable pipeline
- RF overfitted (train R^2 0.97 vs test 0.74, $n=267$)
- Logistic Reg. outperformed ensemble methods

Strengths

- Zero-difference cross-validation between MYEB2 and MYEB3
- All 3 required technique types implemented and compared
- Modular, annotated R pipeline - fully reproducible with `set.seed(42)`

Limitations

- XGBoost classification may reflect a possible encoding/tuning issue
- Small training set ($n=267$) limits ensemble model generalisation
- No spatial modelling - geographic patterns unaccounted for